Generating Individual-Level Full-Day Activity Trajectory via Large Language Models

Yezhen Zhang, Zhengzi Zhou, Yanbo Pang, and Chen Zhong

Centre for Advanced Spatial Analysis, University College London

June 5, 2026
Ghent, Belgium

Contents

- Motivation
- Methodological Challenge
- Framework
- Experiment
- Results
- Conclusion
- Future Work
- Acknowledgements

Motivation

- Human mobility data are valuable, but face two key **challenges**
  - Privacy concerns (eg. GPS and LBS data)
  - Data utility limitations (eg. Travel surveys)

- Synthetic mobility data can provide **alternatives** to raw mobility data while preserving analytical use

Methodological Challenge

- Existing approaches to synthetic mobility data generation face **challenges**.

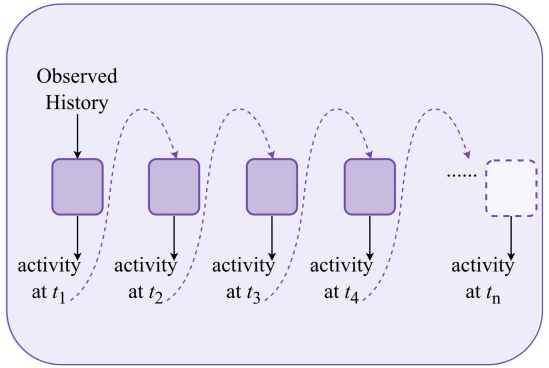
- Non-LLM-based Methods
  - Statistical Models: struggle to capture **long-term dependencies**
  - Deep Learning Models: lack **interpretability**

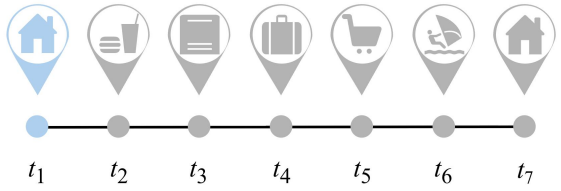
- Adapting general-purpose LLMs to synthetic mobility data generation is non-trivial

Evolution of LLM-based Individual-level Mobility Modelling

(a) Next Location Prediction (Existing)

Modelling Structure (Autoregressive)

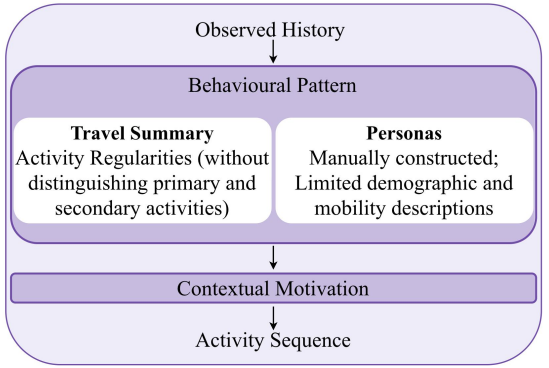


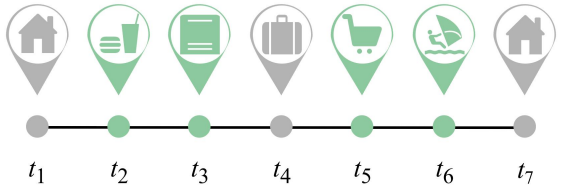


Capability:
Predict only next activity at a time.
Autoregressive approach is required to generate activity sequence.

(b) Activity Sequence Generation (Existing)

Modelling Structure (Hierarchical)

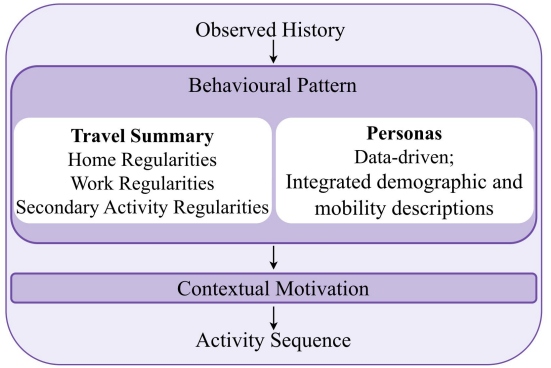


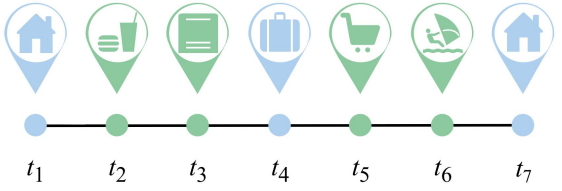


Capability:
Generate activity sequence at a time, without explicitly considering primary activities.

(c) Activity Sequence Generation (Ours)

Modelling Structure (Hierarchical)





Capability:
Generate activity sequence at a time, covering both primary and secondary activities.



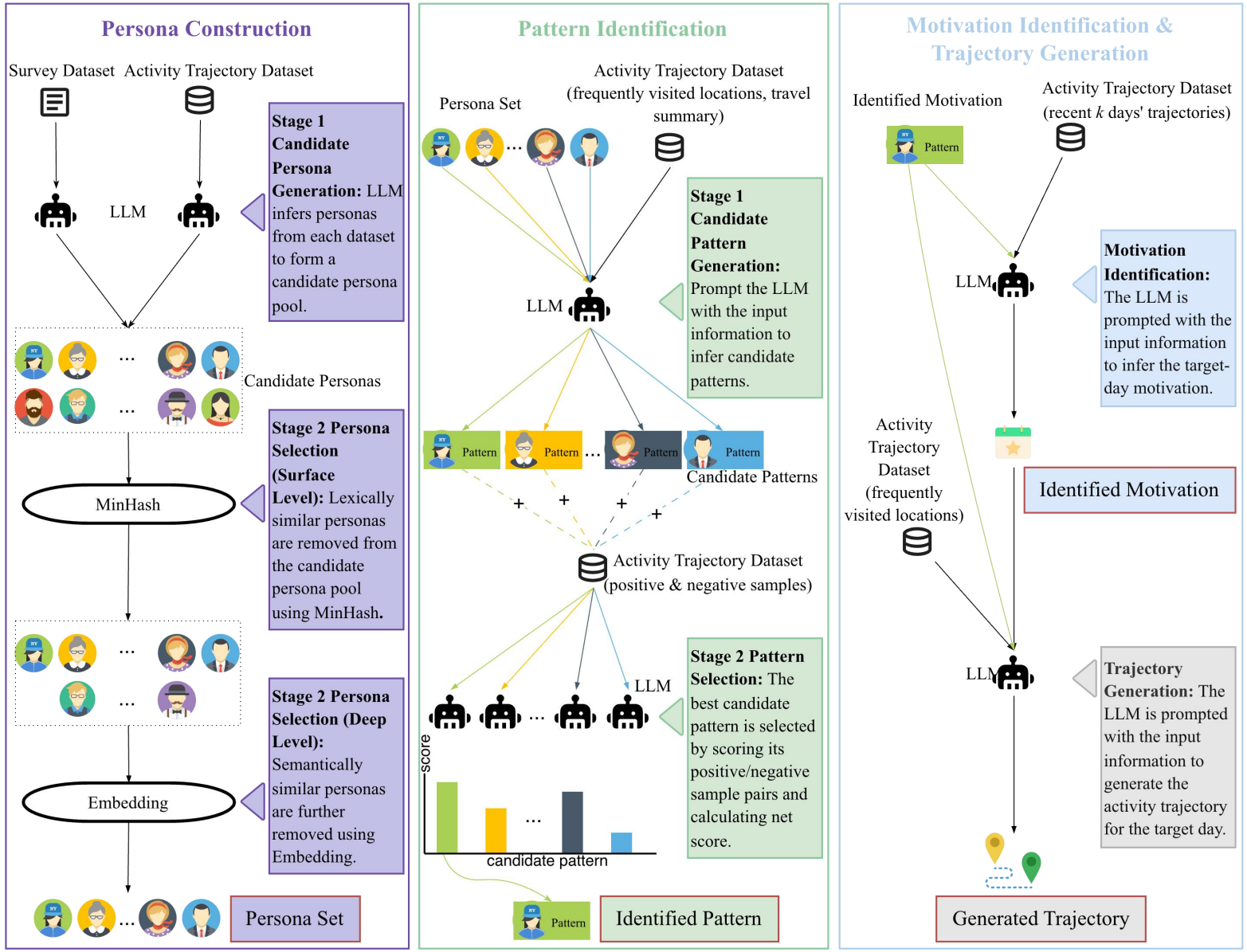
Framework: Preliminaries

- **Task Definition:** Conditional Individual-Level Daily Activity Trajectory Generation
  - Historical trajectories over the past n days → Activity trajectory for the target day $\{(l_0,t_0),(l_1,t_1),\ldots,(l_n,t_n)\}$
  - **Activity:** A stay at a location associated with a specific purpose and time.
  - **Trajectory:** A time-ordered sequence of activities performed by an individual within a day.

Framework: Key Concepts

- Core Generation Mechanism
  - Trajectory = LLM (Pattern, Motivation)

- Additional Prompt Components
  - Travel Summary → structurally complete trajectories
  - Persona → realistic and diverse trajectories


Persona Construction
Survey Dataset
Activity Trajectory Dataset
LLM
Stage 1 Candidate Persona Generation: LLM infers personas from each dataset to form a candidate persona pool.
Candidate Personas
MinHash
Stage 2 Persona Selection (Surface Level): Lexically similar personas are removed from the candidate persona pool using MinHash.
Embedding
Stage 2 Persona Selection (Deep Level): Semantically similar personas are further removed using Embedding.
Persona Set
Pattern Identification
Persona Set
Activity Trajectory Dataset (frequently visited locations, travel summary)
LLM
Stage 1 Candidate Pattern Generation: Prompt the LLM with the input information to infer candidate patterns.
Pattern
Candidate Patterns
Activity Trajectory Dataset (positive & negative samples)
LLM
score
candidate pattern
Stage 2 Pattern Selection: The best candidate pattern is selected by scoring its positive/negative sample pairs and calculating net score.
Pattern
Identified Pattern
Motivation Identification & Trajectory Generation
Identified Motivation
Activity Trajectory Dataset (recent k days' trajectories)
Pattern
LLM
Motivation Identification: The LLM is prompted with the input information to infer the target-day motivation.
Activity Trajectory Dataset (frequently visited locations)
Identified Motivation
LLM
Trajectory Generation: The LLM is prompted with the input information to generate the activity trajectory for the target day.
Generated Trajectory
Persona Set
Pattern
Identified Pattern
Identified Motivation
Generated Trajectory

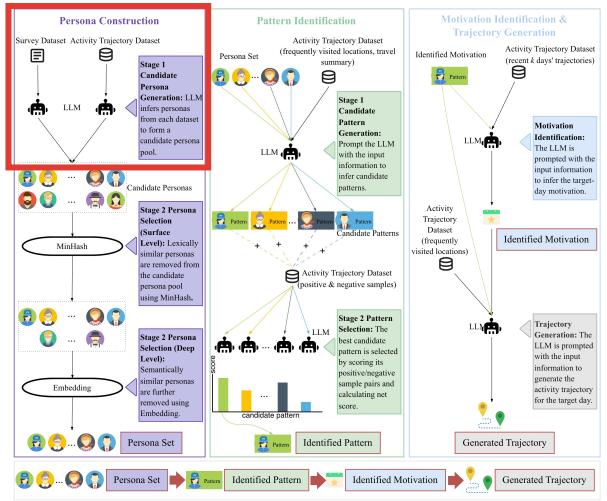
Your task is to generate a persona based on the provided dataset.

You are given a JSON dataset derived from a mobile GPS dataset. It includes:
- A top-level "individual id";
- An "trajectories" array, where each element represents one day and contains that day's sequence of stay-point activities.

Think step by step:
- Read the person's daily trajectories and identify key mobility patterns;
- Based on these mobility patterns, infer plausible socioeconomic characteristics;
- Combine the inferred socioeconomic profile and travel behavior into a single persona.

Each persona must:
- Provide a concise description of this persona (maximum 30 words);
- Provide a role name based on the role description;
- Use professional, neutral language suitable for academic research;
- Follow this exact format:
[Role Name]: [Description of this role.]

Example:
Commuting Professional: Full-time employee, middle age, commuting daily by car, balancing family life with a stable income.



Example prompt template for **persona generation** based solely on mobility trajectories.

Your task is to generate a persona based on the provided dataset.

You are given a JSON dataset derived from the UK National Travel Survey. It includes:
- Top-level "individual id" and "household id";
- An "individual" object describing personal socioeconomic attributes;
- A "household" object describing household socioeconomic attributes;
- A "trip" array, where each element represents one day and contains the list of trips made on that day.

Think step by step:
- Read the person's socioeconomic attributes to form an initial understanding of their demographic and lifestyle background;
- Examine the person's daily trips to identify key travel behaviors;
- Combine the socioeconomic profile and travel behavior into a single persona.

Each persona must:
- Provide a concise description of this persona (maximum 30 words);
- Provide a role name based on the role description;
- Use professional, neutral language suitable for academic research;
- Follow this exact format:
[Role Name]: [Description of this role.]

Example:
Commuting Professional: Full-time employee, middle age, commuting daily by car, balancing family life with a stable income.

Example prompt template for **persona generation** using travel survey data.

Your task is to generate a pattern description that reflects an individual's regular travel behavior based on the given contextual information.

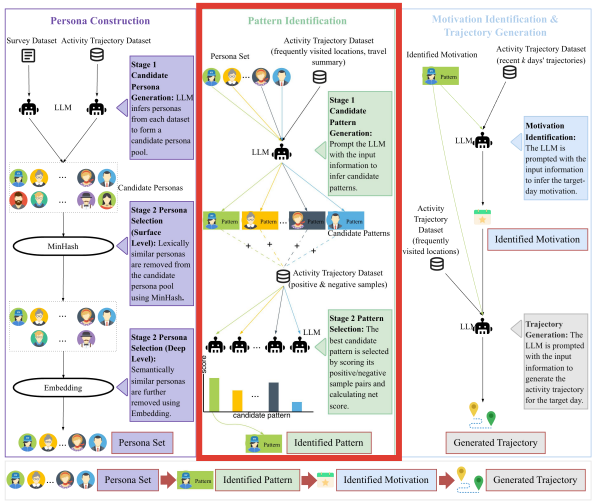
You are provided with the following information:
- Persona: You are a !<INPUT 0>!. !<INPUT 1>!.
- Travel summary: !<INPUT 2>!.
- Additional frequently visited locations: !<INPUT 3>!.

Think step by step:
- Read the persona to understand the individual's background and lifestyle;
- Review the travel summary and the additional frequently visited locations to identify behavioral regularities;
- Based on these insights, describes the individual's typical mobility pattern.

Each output must:
- Remain realistic and internally consistent with the provided persona, travel summary, and locations;
- Consist of one coherent paragraph.

Text to Continue: I am !<INPUT 0>!. !<INPUT 1>!. My typical daily pattern...

Note:
!<INPUT 0>! persona name
!<INPUT 1>! persona description



Example prompt template for **candidate pattern generation**.

Your task is to evaluate how consistent a given daily trajectory is with a specified behavioral pattern.

You are provided with the following information:
- Behavioral pattern: !<INPUT 0>!
- Trajectory sample: !<INPUT 1>!

Think step by step:
- Read and understand the behavioral pattern to identify its core characteristic
- Examine the given trajectory and compare it with the behavioral pattern, focusing on the alignment of activities, temporal order, and spatial context.
- Assign a consistency rating from 1 to 10, and briefly explain your reasoning.

Each output must:
- Provide a numerical rating (1–10) representing the degree of consistency, where 10 = perfectly matches the behavioral pattern, 1 = completely inconsistent with the behavioral pattern.
- Include a concise explanation for the rating (1-2 sentences).
- Follow this exact format:
Rating: <fill in>. Reason: <fill in>

Example:
Rating: 8. Reason: The trajectory closely matches the commuting routine described in the pattern, including similar work hours and activity types. Minor deviations occur in evening leisure locations, but overall consistency is high.

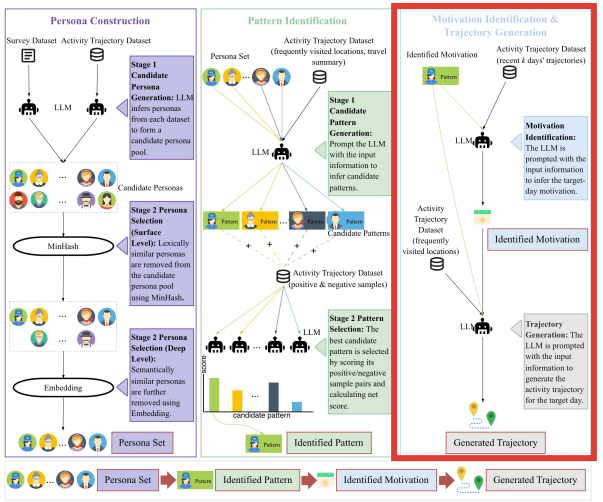
Example prompt template for **pattern selection**.

Your task is to infer the individual's mobility motivation for today based on their established mobility pattern and recent trajectories.

You are provided with the following information:
- Identified mobility pattern: !<INPUT 0>!
- Recent trajectories: !<INPUT 1>!

Think step by step:
- Read the mobility pattern to understand the individual's long-term travel regularities and lifestyle;
- Analyze the recent trajectories to identify short-term deviations, new routines, or contextual factors;
- Integrate both long-term and short-term behavioral information to infer a plausible motivation for today's mobility.

Each output must:
- Provide a concise, realistic description (1–2 sentences) of the inferred motivation;
- Be consistent with both the behavioral pattern and recent trajectories.



Example prompt template for **motivation iidentification**.

Your task is to generate a realistic trajectory plan for today based on the given mobility pattern, current motivation, and frequently visited locations.

You are provided with the following information:
- Mobility pattern: !<INPUT 0>!
- Current motivation: !<INPUT 1>!
- Frequently visited locations: !<INPUT 3>!

Think step by step:
- Review the mobility pattern to understand the individual's long-term routines, anchor points, and mobility tendencies;
- Consider the current motivation and determine how it may influence today's schedule;
- Use the provided frequently visited locations to construct a plausible sequence of activities for today, including times and locations;
- Briefly explain the reasoning behind the generated trajectory.

Each output must:
- Be written strictly in the following JSON format;
{"plan": [<Location> at <Time>, <Location> at <Time>,...]
"reason":...}

Example:
{"plan": ["education#123 at 8:30", "eating and drinking#456 at 11:30", "education#789 at 12:50", "home#405 at 17:20"]
"reason":"My plan today includes attending school in the morning, taking a lunch break, returning to school in the afternoon, and going home afterward."}

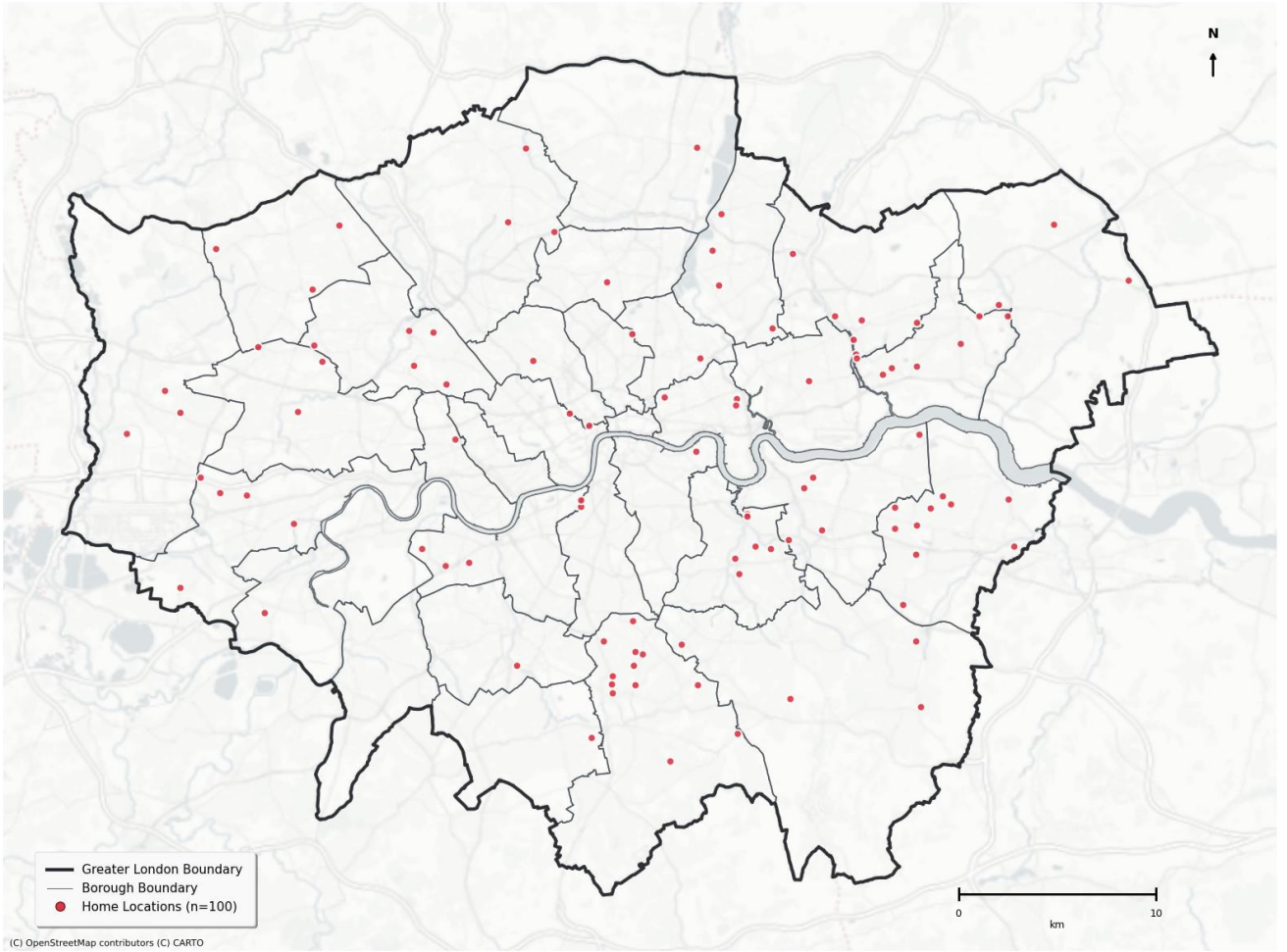
Example prompt template for **trajectory generation**.

Travel Summary Template

Day type	Activity type	Recorded information
Weekday or Weekend	Home	Typical first departure time from home
		Typical final return time to home
		Typical number and timing of returns to home between the first departure and the final return
	Work	Typical number and timing of visits to work locations
	Secondary activities	Typical location and timing of the first secondary activity visit
		Typical location and timing of the last secondary activity visit
		Typical locations and timings of the top-*k* most fre_x0002_quently visited secondary activities

Experiment

- Location: London, UK
- Time: October 2021
- Datasets:
  - Mobile in-app dataset (100 users)
  - UK national travel survey
- Model: GPT-4o-mini
- Baselines:
  - Markov (statistical)
  - LSTM (deep learning)
  - LLMob (LLM)



Results: Personas

Student Commuter: Young individual in the children and teenagers age band, traveling frequently for education, utilizing both car and walking as primary modes of transport.
Childhood Explorer: Active child or teenager, frequently walking and using a private car for social visits and personal errands, from a moderately affluent household.
Young Escort: A child or teenager from a lower-middle-income household, participating in regular short walking trips for escorting family members during personal errands.
Balanced Lifestyle Traveller: This persona strategically balances time between home, work, and leisure activities, with a focus on education and social engagement.
Active Young Adult: Unemployed young adult, cycling several times weekly for personal business, with a low household income.
Home-Based Routine Organizer: This individual maintains a structured daily routine with significant time spent at home, including varied activities such as shopping and work.
Structured Commuter: Middle-aged full-time employee, residing in a high-income household, commuting daily by car, consistently maintaining a structured work-life balance.
Part-Time Senior Walker: Senior individual, working part-time from home, with moderate income; engages in local walking trips and occasional car journeys for shopping and leisure.
Retired Car User: Retired individual, living in a low-income household, reliant on a private car for regular food shopping trips, with minimal use of public transport.

Results: Generative Performance

Models	SD↓	SI↓	DARD↓	STVD↓
Markov	0.0986	0.0078	0.1702	0.6802
LSTM	0.0988	0.1319	0.2729	0.6892
LLMob	0.0195	0.0356	0.1161	0.5297
Ours (w/o *S*)	0.0187	0.0277	0.1098	0.4702
Ours (w/o *P*)	0.0052	0.0030	0.0418	0.2625
Ours	**0.0045**	**0.0021**	**0.0405**	**0.2305**

- Evaluation Metric
  - **JSD** (Jensen–Shannon Divergence)
  - dissimilarity: lower values indicate better alignment
- Mobility Characteristics
  - **SD** (Step Distance): Distance between consecutive activities
  - **SI** (Step Interval): Time gap between consecutive activities
  - **DARD** (Daily Activity Routine Distribution): Joint distribution of activity types and occurrence times
  - **STVD** (Spatial–Temporal Visit Distribution): Joint distribution of activity locations and occurrence times

Results: Ablation Studies

Models	SD↓	SI↓	DARD↓	STVD↓
Markov	0.0986	0.0078	0.1702	0.6802
LSTM	0.0988	0.1319	0.2729	0.6892
LLMob	0.0195	0.0356	0.1161	0.5297
Ours (w/o *S*)	0.0187	0.0277	0.1098	0.4702
Ours (w/o *P*)	0.0052	0.0030	0.0418	0.2625
Ours	**0.0045**	**0.0021**	**0.0405**	**0.2305**

- Evaluation Metric
  - **JSD** (Jensen–Shannon Divergence)
  - dissimilarity: lower values indicate better alignment
- Mobility Characteristics
  - **SD** (Step Distance): Distance between consecutive activities
  - **SI** (Step Interval): Time gap between consecutive activities
  - **DARD** (Daily Activity Routine Distribution): Joint distribution of activity types and occurrence times
  - **STVD** (Spatial–Temporal Visit Distribution): Joint distribution of activity locations and occurrence times

Conclusion

- Propose an **LLM-based framework for conditional individual-level daily activity trajectory generation**, enabling high-fidelity, synthetic alternatives to raw mobility trajectories.
- Introduce a **travel summary** that explicitly models primary and secondary activities, enabling the generation of structurally complete trajectories.
- Develop a **data-driven persona construction pipeline** that enables reproducible and transferable persona generation.
- Demonstrate through experiments on real-world data that the proposed framework achieves **higher fidelity** to real-world mobility characteristics than existing methods.

Future Work

- Improve geopatial representations



- Examine generalisability across diverse contexts
- Conduct formal evaluation of privacy guarantees and validation on downstream tasks

Models	SD↓	SI↓	DARD↓	STVD↓
Markov	0.0986	0.0078	0.1702	0.6802
LSTM	0.0988	0.1319	0.2729	0.6892
LLMob	0.0195	0.0356	0.1161	0.5297
Ours (w/o *S*)	0.0187	0.0277	0.1098	0.4702
Ours (w/o *P*)	0.0052	0.0030	0.0418	0.2625
Ours	**0.0045**	**0.0021**	**0.0405**	**0.2305**

- Evaluation Metric
  - **JSD** (Jensen–Shannon Divergence)
  - dissimilarity: lower values indicate better alignment
- Mobility Characteristics
  - **SD** (Step Distance): Distance between consecutive activities
  - **SI** (Step Interval): Time gap between consecutive activities
  - **DARD** (Daily Activity Routine Distribution): Joint distribution of activity types and occurrence times
  - **STVD** (Spatial–Temporal Visit Distribution): Joint distribution of **activity locations** and occurrence times

Future Work

- Improve geospatial representations



- Examine generalisability across diverse contexts
- Conduct formal evaluation of privacy guarantees and validation on downstream tasks

Acknowledgements

- This project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No 949670), and Proof of Concept (grant agreement No 101212775).

Thank You